The Risk Issue — Artometrics

The Risk Issue

Airlines, storms, nuclear ledgers, hospitals — institutional risk as narrative and number.

Which Airlines Have the Worst Safety Records?

Airline safety is usually discussed one crash at a time. The TidyTuesday airline-safety extract does something different: it lines carriers up by available seat kilometers per week — a capacity metric — and then asks how incidents, fatal accidents, and fatalities sit against that scale. The working file holds **336** cleaned records after the week's tables are merged.

Capacity is not the same as risk, but it is the right denominator for a first cut. A carrier flying **7.1 billion** available seat kilometers per week is not in the same exposure class as one flying under **2 billion**. The median in this extract is **802,908,893** available seat kilometers per week — a calibration point for every ranking that follows.

The most common event type in the file is simply labeled **incidents**. That matters: the archive mixes routine reportable events with rarer fatal outcomes, and treating those categories as interchangeable would flatten the story.

Fast facts

The numbers that set the scale for this report:

336Records in the working dataset

802,908,893Median Avail seat km per week
7,139,291,291Highest observed Avail seat km per week
United / Continental*Top Airline by Avail seat km per week
incidentsMost common Type of event

Data and method

The source is the TidyTuesday release dated 2018-08-07 from the R for Data Science community. After merging the week's CSV and XLSX tables, the working frame has 336 rows and five analysis columns used in these charts.

Medians are preferred wherever available seat kilometers skew toward megacarriers. Index-style fields are excluded from metric selection. Charts ship as Plotly JSON with PNG fallbacks so values remain inspectable on hover.

Capacity Concentrates Among Megacarriers

Avail seat km per week by Airline

United / Continental* anchors the top of the capacity ladder at **7,139,291,291** available seat kilometers per week. Delta / Northwest* follows at about **6.53 billion**, then American* at **5.23 billion**. The next tier — Lufthansa*, Southwest, British Airways*, Air France — sits between roughly **3.0 and 3.4 billion**.

That gap is structural. The top three merged U.S. brands occupy a different scale class from even large European flag carriers in this extract. When safety counts are later compared across airlines, those capacity differences are the first context any fair reading needs.

The Same Names Dominate the Leader Board

United / Continental * leads at 7,139,291,291 — 3,091,881,806 marks the median among the top dozen

The leaders chart restates the capacity hierarchy with a sharper claim: United / Continental* leads at **7,139,291,291**, while the median among the top dozen sits near **3,091,881,806**. Half the “elite” group is still less than half as large as the leader.

In other words, “large airline” is not a single club. It is a steep pyramid. Any narrative that ranks raw incident counts without that pyramid will punish the biggest networks for existing.

Event Type Does Not Rewrite the Capacity Story

Avail seat km per week by Type of event

Box plots split available seat kilometers by event type — incidents, fatal_accidents, and fatalities. Each group shows the same median of **802,908,893** and the same max of **7,139,291,291** across 112 observations per box in this view.

That congruence is the finding. Event labels do not sort carriers into different capacity worlds; the same megacarrier scale appears whether the row is tagged as an incident or a fatality record. The distribution is about who flies a lot, not about which label the row carries.

Gaps to the Median Are Flat Across Labels

Avail seat km per week vs median by Type of event

When available seat kilometers are compared to the sample median by event type, incidents, fatal_accidents, and fatalities all land at essentially zero gap in this chart. That is a stability check, not a dramatic ranking.

The chart’s job is to stop a false story: that one event category is concentrated among unusually large or unusually small carriers. In this extract, the labels do not pull capacity away from the median.

Scale and Event Counts Do Not Move as One Line

Avail seat km per week vs N events

Plotting available seat kilometers against the number of events reveals clusters the averages erase. Some mid-size carriers post high incident counts; some of the largest networks sit closer to moderate event totals relative to their seat-kilometer footprint.

The fatality panel is especially uneven: the highest fatality totals in the scatter are not always the absolute capacity leaders. Exposure and outcome are related, but they are not a straight diagonal — which is exactly why capacity and counts must be read together.

What this file cannot tell you

This is a community-cleaned TidyTuesday snapshot, not a live regulatory feed. Missing values, spelling variants on airline names (including starred merger labels), and week-of-export coverage limits all apply. Merged tables can duplicate or fan out rows when join keys are imperfect.

Available seat kilometers measure exposure opportunity, not proven risk. The file cannot adjudicate which carriers are "safest" in a causal sense; it can only show how capacity and event tallies sit side by side in this extract.

What to take away

The clearest signal is structural: a handful of megacarriers dominate available seat kilometers, with United / Continental* at the top and a median near **803 million** seat kilometers per week for the wider field.

Event-type labels do not reorder that hierarchy. The useful question is not which airline has the most rows — it is how those rows scale against who actually flies the world's seat kilometers.

Sources

Data Science Learning Community. (2018). *TidyTuesday: Airline Safety*. https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2018/2018-08-07/week19_airline_safety.csv

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Which Storms Hit Puerto Rico Hardest?

In late summer 2017, the Atlantic hurricane season forced a brutal comparative question into public view: which places absorbed the worst of the storm window, and how did Puerto Rico's readings sit beside Florida and Texas?

A compact TidyTuesday file of **153** records tracks values across that season's geography. The median reading is **703**; the highest observed value hits **5,072**. Texas leads the state-level totals in the working extract, but the timeline chart is where Puerto Rico's distinct peak pattern becomes visible.

This is not a full climatology of Caribbean cyclones. It is a season-slice that lets three hard-hit places be read side by side — intensity, concentration, and timing included.

Fast facts

The numbers that set the scale for this report:

153Records in the working dataset
703Median Value
5,072Highest observed Value
TexasTop State by Value
2017–2017Year span covered in the file

Data and method

The source is the TidyTuesday release from 2018-06-19 (R for Data Science community). The working file contains 153 rows and 4 columns after cleaning — a narrow table built for comparison across states and territories in the 2017 window.

Medians are preferred for robustness. Charts ship as Plotly JSON with PNG fallbacks. Because the year span in the extract collapses to **2017–2017**, conclusions are about that season's comparative geography, not about multi-decade hurricane climatology.

Timeline across states

Value by State

Daily value lines separate which state or territory bore the brunt on which days. Peaks rarely align. Texas, Florida, and Puerto Rico experience the same basin season on different clocks — landfall timing and local intensity create staggered curves rather than a single shared crest.

The maximum reading of **5,072** defines the intensity ceiling for the window. Everything else in the file is best read relative to that spike and to the median of 703.

Who sits at the top

Texas leads at 983 — 621 marks the median among the top dozen

On the leader ranking used here, **Texas** leads at **983**, with **621** as the median among the top dozen entries. Florida and Puerto Rico complete the short list of places that dominate the aggregate signal.

A short leaderboard is itself a finding: in this file, storm impact is not evenly sprinkled across dozens of states. A handful of geographies carry the measurable season.

How values are spread

Median 703 vs mean 1,020 — the shape is right-skewed

The distribution is right-skewed: median **703** versus mean **1,020**. The top decile begins near **2,214**. That tail is where the defining storm days live — and why a mean-based summary will sound more extreme than a typical day in the file.

Skewness is the statistical signature of disaster data. Most observations are elevated but survivable; a minority of days rewrite the season's memory.

Concentration of impact

The top 3 state entries account for 100% of the aggregate value

In this working extract, the top three state or territory entries account for **100%** of the aggregate value. The Pareto curve is not merely steep — it is effectively a three-place map.

That concentration is why comparative journalism about 2017 kept returning to Texas, Florida, and Puerto Rico. The file's structure matches the public memory: a short head, not a long democratic list of mildly affected places.

Concentration, from another cut

The top 3 state entries account for 100% of the aggregate value

A second concentration chart repeats the same structural claim from a related cut of the table: the measurable aggregate is carried by the same three-place head. Steep Pareto curves mean a small head drives most of the signal.

For Puerto Rico specifically, the lesson is not that other places did not suffer. It is that in this season-slice, comparative magnitude collapses onto a tiny set of names — and Puerto Rico is inside that set.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live APIs. Missing values, spelling variants, and week-of-export coverage limits apply. A 153-row season extract cannot stand in for FEMA damage ledgers, mortality studies, or multi-decadal storm catalogs.

Value in the working columns should be read as the file's measured intensity metric — not as a universal currency of loss. Pair these charts with official after-action reporting before converting them into legal or fiscal claims.

What to take away

The 2017 window in this file is a story of concentration: median 703, ceiling 5,072, and an aggregate dominated by three places. Texas leads the ranking cut; Puerto Rico remains essential to the timeline.

Disaster seasons are not democratic distributions. They are skewed clocks. The citable map is the short head of places where the season actually happened.

Sources

How Nuclear Test Yields Shifted Through the Atomic Age

Nuclear test histories are usually told as geopolitics. The event catalogs underneath — body-wave magnitudes, dates, purposes, and depths — also form a quantitative archive of how intensely states practiced the bomb.

A TidyTuesday working extract of **2,051** records puts median body-wave magnitude at **0.00** and the highest observed magnitude at **7.40**. The **USSR** leads the country ranking by magnitude body in the fact box, and purpose code **WR** is the most common label. The year marker in the extract collapses to **1970** for one of the summary spans, so timeline charts should be read as the file's available event clock rather than a claim that testing happened only in a single year.

Zeros in the magnitude field are not moral quiet. They are measurement gaps, non-detections, or unfilled cells that pull the median to the floor while a thin upper tail records the events that shook instruments hardest.

Fast facts

The numbers that set the scale for this report:

2,051Records in the working dataset
0.00Median Magnitude body
7.40Highest observed Magnitude body
USSRTop Country by Magnitude body
1970–1970Year span covered in the file
WRMost common Purpose

Data and method

The source is the TidyTuesday nuclear explosions release maintained by the R for Data Science community. The working file assembles event-level rows with country labels, purpose codes, body-wave magnitude, depth, and related fields after cleaning.

Medians are especially important here because missing and zero magnitudes dominate the center of the distribution. Charts export as Plotly JSON with PNG fallbacks. Public test catalogs are incomplete by nature; secrecy and uneven seismology both bite.

How magnitudes move through the record

Median Magnitude body Over Time

The time chart of median body-wave magnitude tracks how the typical recorded event sits across the file's date spine. Periods with denser instrumentation and more reported yields look different from periods dominated by missing magnitude cells.

Read the curve as a history of what entered the public measurement record — not as a complete energy ledger of every device. Diplomatic moratoria and underground testing both change what seismology sees.

Who sits at the top

INDIA leads at 5.00 — 2.50 marks the median among the top dozen

On the highlighted leader cut, **INDIA** leads at **5.00**, with **2.50** as the median among the top dozen country entries. Related state labels in the broader ranking — USSR, China, Pakistan, France, USA, UK — remind readers that the nuclear club is small and that magnitude leadership in a cleaned table can reflect which events were well measured as much as which states tested most often.

The USSR's fact-box lead by magnitude body and India's lead on this chart can both be true under different aggregations. Always name the cut before converting a bar into a hierarchy of arsenals.

Purpose spreads magnitudes

Magnitude body by Purpose

Box plots by purpose code show whether magnitude consensus is shared or contested across test categories. **WR** dominates headcount as the most common purpose label; other purpose codes may sit higher or lower on intensity even with fewer rows.

Purpose is the political grammar of the catalog: weapons-related tests, peaceful nuclear explosion labels, and related codes are not interchangeable events even when they share an explosive physics.

Who beats the median — and who trails

Magnitude body vs median by Purpose

The gap chart ranks purpose categories above or below the magnitude median. Categories that beat the median concentrate the instrumentally loud events; categories that trail are either genuinely smaller or more missingness-heavy.

Because the file-wide median is 0.00, almost any filled magnitude will sit above the mathematical center. That is a data-structure warning as much as a physics finding.

Magnitude and depth

Magnitude body vs Depth

Plotting body-wave magnitude against depth shows clusters that averages erase. Atmospheric and surface events occupy different depth regimes from underground tests; the scatter is where those testing modes become visible.

Depth is not a perfect proxy for political intent, but it is one of the cleanest engineering signatures in the archive. Combined with magnitude, it separates theater from concealment more clearly than country labels alone.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live APIs. Missing values, spelling variants, and incomplete yield reporting apply. A median magnitude of 0.00 is a missingness signal as much as a physical one.

Findings describe structural signals in the public nuclear-test table — not a classified inventory, not a health study, and not a complete history of every device.

What to take away

The nuclear test archive in this file is a skewed measurement story: 2,051 records, a median body-wave magnitude at zero because of gaps, and a thin upper tail that reaches 7.40.

The citable lesson is double: a small set of states dominates the known record, and the record itself is shaped by what seismology and disclosure were willing to make countable.

Sources

Who Generates the Most Plastic Waste—and Who Mismanages It?

Plastic waste is usually discussed as a moral story about litter. The data make it an economic geography story: who generates waste, who mismanages it, and how those patterns sit against income.

A TidyTuesday compilation spanning country-year records puts **22,204** rows on the table. Median GDP per capita in the working file is **8,447** (PPP, constant 2011 international dollars). The highest observed rate reaches **135,319**, with Macao at the top of that income ranking. The year span runs from **1700** to **2017** — long enough to watch modern prosperity and modern waste arrive together.

Income alone does not settle the mismanagement question. Rich places can fund collection systems; poorer coastal places can become leakage hotspots even at lower per-person generation. The charts below separate those layers.

Fast facts

The numbers that set the scale for this report:

22,204Records in the working dataset
8,447Median GDP per capita, PPP (constant 2011 international $) (Rate)
135,319Highest observed GDP per capita, PPP (constant 2011 international $) (Rate)

MacaoTop Entity by GDP per capita, PPP (constant 2011 international $) (Rate)
1700–2017Year span covered in the file

Data and method

The source is the TidyTuesday release from 2019-05-21 (R for Data Science community). The working file contains 22,204 rows and 7 columns after assembling the week's tables — including GDP per capita (PPP) and per-capita mismanaged plastic waste in kilograms per person per day.

Medians are used where distributions skew. Charts export as Plotly JSON with PNG fallbacks. Country names, missing years, and sparse early coverage mean the long historical span is denser for income than for waste metrics; read early centuries as context for prosperity, not as a continuous waste census.

How prosperity moved over time

Median GDP per capita, PPP (constant 2011 international $) (Rate) Over Time

Median GDP per capita rises from roughly **5,891** in the opening period of the series to about **12,592** at the close. That climb is the backdrop for plastic: mass polymers are a late-twentieth-century material living inside an income distribution that was already diverging for centuries.

The time chart is not a plastic-generation curve by itself. It is the economic stage on which generation and mismanagement later play out — and it reminds readers that global medians hide enormous cross-country gaps in the same year.

Who sits at the top of income

Qatar leads at 118,396 — 67,799 marks the median among the top dozen

Among the highest GDP-per-capita observations, **Qatar** leads near **118,396**. The median among the top dozen sits around **67,799** — still many times the file-wide median of 8,447. Macao, Singapore, Switzerland, and other high-income city-states and petro-economies fill the same thin upper band.

These leaders matter for plastic debates because high income correlates with high consumption — but not automatically with high mismanagement. Collection infrastructure can rise with prosperity even as packaging intensity rises with it.

How income is spread

GDP per capita, PPP (constant 2011 international $) (Rate) Distribution

The distribution is right-skewed: median **8,447** versus mean **14,926**. The top decile begins near **37,969**. That tail is where defining high-income cases live — and where mean-based storytelling will overstate how typical prosperity feels.

Most country-year observations sit far below the headline petro-state and city-state peaks. Any claim about the world and plastic that starts from the mean income will describe a richer planet than the median resident actually inhabits.

Leader trends

Top Entity Over Time

The leading names do not move in lockstep. Some high-income series fade as others surge; oil-price cycles and small-population city economies can reshuffle the top of a PPP ranking without changing the deeper structure of global waste.

Tracking leaders over time separates sustained dominance from one-off spikes. For plastic policy, sustained high-income systems with strong waste services are a different object than briefly elevated GDP readings in small jurisdictions.

Income versus mismanaged plastic

GDP per capita, PPP (constant 2011 international $) (Rate) vs Per capita mismanaged plastic waste (kilograms per person per day)

Plotting GDP per capita against per-capita mismanaged plastic waste shows clusters that averages erase. Some rich economies sit low on mismanagement — consumption with collection. Some lower-income coastal economies sit high on mismanagement even when generation per person is modest.

That scatter is the core policy map. Generation without management becomes ocean-bound leakage; management without reduced generation becomes an expensive success. The file cannot settle every causal claim, but it can stop the false equation that wealth alone explains visible plastic pollution.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live APIs. Missing values, spelling variants, and week-of-export coverage limits apply. Early historical GDP rows are not paired with modern waste instrumentation.

Mismanaged waste estimates depend on modeling assumptions about litter, dumping, and collection coverage. Treat the relationship charts as structural signals about income and leakage risk — not as a courtroom inventory of every country's shoreline.

What to take away

Plastic waste sits at the intersection of prosperity and infrastructure. Median incomes in the file roughly double over the long span, while mismanagement risk remains geographically uneven.

The citable lesson is modest and sharp: high GDP per capita identifies capacity, not automatic cleanliness, and low GDP per capita identifies constraint, not automatic innocence. The scatter of income against mismanagement is where those two truths become visible at once.

Where Wastewater Plants Serve the Most People

Wastewater plants are the unglamorous infrastructure of modern density. They turn the output of cities into something rivers and coasts can sometimes absorb — and the HydroWASTE-style plant register makes that system countable.

The working TidyTuesday extract holds **58,502** records. Median waste discharge (**WASTE DIS**) sits at **1,079**; the highest observed value exceeds **3,073,754**. The United States leads on aggregate discharge in the country rollup, while **Secondary** treatment is the most common level label in the file.

Scale without treatment level is an incomplete story. A large plant with advanced treatment is a different civic object from a large plant stuck at primary or secondary standards.

Fast facts

The numbers that set the scale for this report:

58,502Records in the working dataset
1,079Median WASTE DIS
3,073,754Highest observed WASTE DIS
United StatesTop COUNTRY by WASTE DIS
SecondaryMost common LEVEL

Data and method

The source is the TidyTuesday release from 2022-09-20 (R for Data Science community). The working file contains 58,502 rows and 25 columns after merging available tables — country, treatment level, discharge, and related quality fields among them.

Medians handle skew from a handful of enormous plants. Charts export as Plotly JSON with PNG fallbacks. Status labels such as operational, proposed, or decommissioned matter for interpretation: not every row is an active civic asset.

Discharge by country

WASTE DIS by COUNTRY

Country rollups of waste discharge put high-intensity systems in view. In the chart's highlighted comparison band, **Singapore** leads near **223,683** on the displayed metric while **Nicaragua** anchors a lower end near **33,649** — a reminder that country averages mix plant size, urban concentration, and reporting coverage.

The United States leads the fact-box country ranking by aggregate discharge, which is a different statement from per-plant intensity. Large federations with many recorded plants will dominate totals even when median plants elsewhere look bigger.

Who sits at the top

Singapore leads at 223,683 — 56,849 marks the median among the top dozen

Among the top dozen country entries on the intensity cut, Singapore leads at **223,683**, and the median of that elite group is **56,849** — far above the file-wide median of 1,079.

Dense city-states and highly instrumented reporting systems rise on these lists. The ranking is as much about what gets measured and attributed as about what flows through pipes.

Treatment levels spread discharge

WASTE DIS by LEVEL

Category boxes by treatment level — secondary, advanced, and related labels — show whether discharge consensus is shared or contested across tiers. Secondary dominates headcount; advanced plants can dominate intensity.

That split is the civic stakes of the chart. Upgrading level without changing plant counts can move environmental outcomes more than building another secondary facility of median size.

Who beats the median — and who trails

WASTE DIS vs median by LEVEL

Advanced treatment sits **681** above the median on the gap chart; **Secondary** trails by **342**. The direction matches intuition only if you remember that advanced plants are often larger urban facilities, not merely cleaner ones.

Level labels encode technology and, often, city scale. Reading the gap as a pure virtue ranking of treatment quality without plant-size context will overfit the moral and underfit the engineering.

Discharge and quality fields

WASTE DIS vs QUAL WASTE

Plotting waste discharge against the file's quality-related waste field shows clusters that averages erase. Some plants combine high discharge with strong quality scores; others pair modest discharge with weaker quality flags.

The scatter is where big and good stop being synonyms. Infrastructure policy needs both axes: capacity to serve population, and treatment quality sufficient for the receiving water.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live APIs. Missing values, spelling variants, and uneven national reporting apply. Proposed or non-operational plants can sit beside active ones if status filters are imperfect.

Discharge units and quality fields should be read as the HydroWASTE-derived attributes in the release — not as a substitute for local permit databases or real-time effluent monitoring.

What to take away

Wastewater is a skewed industry: median discharge near 1,079, extremes above three million, and country totals dominated by large reporting systems such as the United States.

The citable distinction is treatment level versus size. Secondary is common; advanced sits above the median on intensity; quality and discharge together decide whether a plant is merely large or actually protective.

Sources

Which Hospitals Still Fail the 30-Day Readmission Standard?

The Hospital Readmissions Reduction Program is a Medicare penalty system that has been running since 2012. The premise is simple: if your patients come back to the hospital within 30 days of discharge at a higher rate than expected given their medical profile, CMS docks your Medicare reimbursements. The penalty caps at 3% of all Medicare payments — not just payments for the specific condition being measured. A hospital with too many heart failure readmissions doesn't just lose heart failure revenue. It loses 3% of everything.

The core metric is the Excess Readmission Ratio, or ERR. It is a ratio of predicted readmissions to expected readmissions — where "expected" is risk-adjusted for each patient's age, comorbidities, and discharge history. An ERR of exactly 1.0 means you readmitted exactly as many patients as a hospital of your type and patient mix should. Above 1.0 means more readmissions than expected. Below 1.0 means fewer. What this analysis found: nearly every condition's national average ERR is above 1.0. HRRP has been running for over a decade, and the average American hospital is still readmitting more patients than CMS expects. The penalty hasn't eliminated the problem. It has clarified it.

Research question

Where do CMS Hospital Readmissions Reduction Program penalties still concentrate — by state, clinical condition, and hospital ownership — in the FY2025 supplemental extract, and what does that geography imply for the 742 hospitals entering CMS's Transforming Episode Accountability Model (TEAM) on January 1, 2026?

This is an observational audit of published CMS ratios, not a causal trial. We do not estimate a formal null hypothesis. We ask a sharper operational question: after more than a decade of HRRP, which named markets (New Jersey, Massachusetts, Mississippi), conditions (Hip/Knee arthroplasty), and ownership types (for-profit systems such as HCA and Tenet) still sit farthest above CMS's expected readmission line?

Fast facts

The numbers that set the scale for this report:

48.1% of HRRP-eligible hospital-condition pairs carried an excess readmission ratio above 1.0 — more readmissions than CMS models predict
6 conditions tracked under HRRP — AMI, Heart Failure, Pneumonia, COPD, Hip/Knee, and CABG
1.0018 average excess readmission ratio nationally — the average hospital readmits slightly more patients than CMS models expect
1.00485 Hip/Knee average ERR — the highest of the six conditions, nearly 2× the excess of the next closest track
742 hospitals mandated to participate in CMS's TEAM bundled payment model as of January 1, 2026 — where readmission penalties become episode-level accountability
65.4% New Jersey's share of penalized hospital-condition pairs — the highest-risk geography in the working dataset

Data and method

The FY2025 HRRP supplemental extract covers the measurement window used for CMS's published excess readmission ratios. After removing suppressed rows — hospitals with fewer than 25 discharges for a given condition, which CMS redacts to protect patient privacy — the working dataset contains 11,720 hospital-condition pairs across roughly 2,700 hospitals and six conditions: Acute Myocardial Infarction (AMI), Heart Failure, Pneumonia, Chronic Obstructive Pulmonary Disease (COPD), Hip & Knee Arthroplasty, and Coronary Artery Bypass Graft (CABG). A hospital can perform well on five tracks and still take a hit on one.

The suppressed rows matter. Small rural hospitals disproportionately fall below the 25-discharge threshold and disappear from the analysis. The hospitals in this dataset skew toward larger, busier facilities. That is not a flaw in the extract — it is a feature of how CMS designed the program — but it shapes every finding below.

An ERR above 1.0 feeds a penalty calculation; the tier depends on how far above 1.0 a hospital sits relative to peers. For this piece, pairs are grouped into four tiers: No Penalty (ERR ≤ 1.0), Low, Medium, and High. Charts are rendered in R from the working CSVs in this monorepo (articles/readmitted/data/), with PNG and interactive Plotly JSON exported side by side. Ownership labels for Chart 3 were joined from CMS Hospital General Information (dataset xubh-q36u) and stored offline so the analysis does not depend on a live CMS download at render time.

The Geography of Failure

States above the national average for penalized hospital-condition pairs — New Jersey leads at 65.4%

> Above the 48.1% national line, New Jersey leads at 65.4% — and high-resource markets sit beside high-poverty ones

Nearly half of all hospital-condition pairs in this dataset carry an excess readmission ratio above 1.0. That national average hides the real story: the penalties are not evenly distributed.

New Jersey leads the country at 65.4% — nearly two of every three hospital-condition pairs in the state perform worse than CMS models expect. Massachusetts (62.5%) and Mississippi (61.1%) follow. New Jersey hosts high-revenue systems such as Hackensack Meridian and RWJBarnabas; Massachusetts concentrates academic medical power around Mass General Brigham; Mississippi is rural, high-poverty, and chronically under-served. Different systems, different patient mixes, same penalty outcome.

Georgia, Kentucky, West Virginia, Alabama, and Louisiana cluster high — which fits the familiar population-health narrative. So do Illinois, California, New York, and Pennsylvania. This is not a rural-poverty map with a clean boundary. High-resource markets do not immunize a system.

Chart 1 shows only states above the national average — the cohort pulling the 48.1% mean up. Even below-average states are not clean. “Below average” means a smaller share of pairs exceed ERR 1.0, not zero penalties.

The Condition Nobody Is Solving

Average excess readmission ratio by HRRP condition — Hip/Knee leads at 1.00485

> All six conditions average above ERR 1.0; Hip/Knee sits at 1.00485 — nearly twice the excess of the next track

All six conditions tracked under HRRP carry an average excess readmission ratio above 1.0. One stands apart.

Every tracked condition still averages above 1.0. The national mean ERR is 1.0018 . Readmission rates have declined since 2012, but the program has not eliminated excess returns — it has measured them.

Hip and knee replacement patients are readmitted at nearly twice the excess rate of the next closest condition. Hip/Knee’s average ERR of 1.00485 sits above COPD (1.00271), Heart Failure (1.00254), AMI (1.00212), Pneumonia (1.00198), and CABG (1.00187). The gaps live in the fourth decimal place — and still matter when multiplied across roughly 1.3 million U.S. joint replacements a year. Elective joint surgery shifts the danger window into post-discharge days hospitals cannot monitor as stays shorten.

COPD sits near the bottom of this chart, but that is not a success story. COPD patients return at high rates in absolute terms. The lower ERR means CMS's risk model expects a sicker baseline. The bar is lower because the patients are harder — not because outcomes are better.

Ownership, Penalty, and Who Pays

Penalty tier distribution by hospital ownership — for-profits carry more high-tier weight

> For-profit hospitals carry more medium and high penalty weight; every ownership type still has a majority in the no-penalty band

For-profit hospitals carry a higher share of medium and high penalty tiers than either non-profits or government hospitals. The policy debate often oversimplifies who is actually paying.

Every ownership type has roughly half or more of its hospital-condition pairs at or below the CMS expected readmission rate. HRRP is often described as a blanket stick. The data says the stick is concentrated — closer to an audit sampling than a universal fine. Systems such as HCA, Tenet, and Steward sit in the for-profit column; that does not mean every for-profit campus is High-tier.

The gap is real — on the order of 8 to 10 percentage points more penalized pairs than non-profits — but it is a difference of degree, not kind. All three ownership types are playing the same game. For-profits lose it slightly more often.

Government-owned facilities land between non-profit and for-profit. Neither the safety-net excuse nor the public-accountability story fits cleanly. County systems operate under political oversight that can slow discharge innovation; middle-of-pack performance is its own finding.

Intensity, Not Just Share

Average excess readmission ratio by state — Massachusetts leads the top 15

> Penalty share and average ERR are related but not identical — Massachusetts leads on intensity even where New Jersey leads on breadth

Chart 1 measured how widely penalties spread across hospital–condition pairs. Chart 4 measures how far above the CMS expected line those pairs sit on average. Massachusetts posts the highest average ERR in the working extract at **1.0344**, ahead of New Jersey's breadth-first pattern at **65.4%** of pairs penalized.

That distinction matters for systems such as Mass General Brigham, Beth Israel Deaconess Medical Center, and the Boston academic corridor: a market can look merely "above average" on share and still run hotter on intensity. Mississippi, Illinois, and several Mid-Atlantic states also appear in the top-15 intensity list — the same mixed geography of high-resource and high-poverty markets seen in Chart 1.

The dotted reference at ERR = 1.0 is the CMS expected line. Every state in this top-15 panel sits above it. The national mean ERR of **1.0018** looks small in isolation; stacked across thousands of pairs and roughly 1.3 million U.S. joint replacements a year, fourth-decimal gaps become budget lines.

Where the High Tier Concentrates

High-tier penalty share by ownership — for-profits at 39.5%

> For-profit hospitals place 39.5% of pairs in the High penalty tier, versus about 32% for non-profit and government peers

Chart 3 showed the full tier mix. Chart 5 isolates the High tier — the pairs farthest above CMS's expected readmission rate. For-profit ownership places **39.5%** of hospital–condition pairs in High, compared with roughly **32.1%** for non-profits and **32.0%** for government facilities.

Named for-profit operators in the CMS Hospital General Information join — including campuses associated with HCA Healthcare, Tenet Healthcare, and historically Steward Health Care — sit inside that column. The chart does not prove that for-profit status causes excess readmissions. It shows that when CMS's published ratios are stacked by ownership, the High-tier mass sits heavier on the for-profit side.

For TEAM's 742 mandatory hospitals in 2026, that concentration is the practical signal: episode-level accountability will land hardest where High-tier HRRP exposure already clusters, not where the "no penalty" majority still holds.

What this file cannot tell you

CMS suppresses readmission data for hospitals that fall below 25 discharges per condition per measurement period. That threshold protects statistical reliability, but the effect is systematic — small rural hospitals disappear from this analysis. The hospitals remaining skew toward larger, busier facilities.

Ownership classification for Chart 3 comes from CMS Hospital General Information, which uses inconsistent labeling across hospital types. Physician-owned facilities, tribal hospitals, and church-affiliated systems do not always map cleanly into three buckets. The Government, Non-Profit, and For-Profit groupings are reasonable approximations — not clean legal categories.

ERR compares actual readmissions to what CMS's risk model predicts for a hospital's patient mix. Case-mix adjustment is imperfect. Hospitals in high-poverty, high-comorbidity markets may still face structural disadvantage the model does not fully absorb. Observation-status shifts after the ACA (documented by Zuckerman et al., 2016) also mean some returns never appear as inpatient readmissions.

What to take away

Nearly half of all hospital-condition pairs still exceed CMS's expected readmission rate — but the distribution is uneven. The highest-penalty states cut across market type and regional demographics. Hip/Knee replacement sits nearly 2× above the next closest condition on excess readmissions. For-profit hospitals carry modestly more penalty exposure than non-profits, though all three ownership types remain majority no-penalty.

HRRP has been in effect since 2012. That 48.1% of pairs still clear the 1.0 ERR threshold more than a decade later says something about the limits of financial penalties as a behavior-change mechanism. Hospitals have responded — rates have declined — but the program has defined the problem more than it has solved it.

CMS's TEAM model, mandatory for 742 hospitals starting January 1, 2026, goes further by tying entire episodes of care to reimbursement. The hospitals that struggled under HRRP are the ones most likely to feel TEAM. This report describes the problem TEAM is designed to address. Whether bundled payments succeed where readmission penalties have not is the next question.

Sources

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Centers for Medicare & Medicaid Services. *Hospital General Information*. CMS Provider Data Catalog, Dataset ID: xubh-q36u. https://data.cms.gov/provider-data/dataset/xubh-q36u — joined to HRRP data for ownership classification in Chart 3.

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Krumholz, H.M., et al. (2017). Relationship between hospital readmission and mortality rates for patients hospitalized with acute myocardial infarction, heart failure, or pneumonia. *JAMA*, 318(3), 270–278. Background on readmission–mortality tradeoffs under quality measurement.

Files

Download the working datasets and chart exports used in this report. All files are hosted on this site — no external repo required.

Datasets

- hrrp_state_summary.csv — state-level penalty share and average ERR
- hrrp_condition_err.csv — average ERR by HRRP condition
- hrrp_ownership_condition.csv — ownership × condition × penalty tier counts

Charts (PNG)

- chart1_states_penalized.png
- chart2_err_by_condition.png
- chart3_penalty_by_ownership.png
- chart4_state_avg_err.png
- chart5_high_penalty_by_ownership.png

Charts (Plotly JSON)

- chart1_states_penalized.plotly.json
- chart2_err_by_condition.plotly.json
- chart3_penalty_by_ownership.plotly.json
- chart4_state_avg_err.plotly.json
- chart5_high_penalty_by_ownership.plotly.json

Editor's note

This report was researched, written, designed, and produced in active collaboration with Claude AI (Anthropic). The data pipeline, statistical analysis, chart design, written analysis, narrative structure, and visual styling were all developed through a directed partnership between human editorial judgment and AI execution. Charts are exported from R in this monorepo (`articles/readmitted/`). Artometrics was built on the premise that rigorous analysis and honest process are not in conflict. The research questions, editorial instincts, interpretive framing, and brand vision are ours. We document this not as a disclaimer but as a description of how we actually work.

— *Artometrics Editorial*

How Human Life Expectancy Doubled in Five Centuries

For most of recorded history, life expectancy at birth hovered between 25 and 40 years — not because adults aged quickly, but because childhood death was common enough to pull the average down. The long arc from that world to the present is one of the few demographic stories that can be told with a single curve and still remain true.

The headline number is staggering: the global median life expectancy nearly triples over five centuries, rising from around 30 to above 73. But the aggregate trend conceals the most important story — the persistent gap between the long-lived and the short-lived. Countries leading this dataset in 2015 live more than twice as long, on average, as the laggards in the same file.

Start with the scale: **73.3** — median life expectancy in 2015, the closing year of the file; and **83.8** — the highest single observation, Hong Kong near peak. The span runs **472 years**, from 1543 to 2015.

Fast facts

The numbers that set the scale for this report:

73.3Median life expectancy in 2015 — the closing year of the dataset
83.8Highest single observation — Hong Kong near peak
33.9→73Median lifespan: opening year to closing year
Hong KongHighest average life expectancy over the full dataset
472 yrsTotal span of the dataset: 1543 to 2015

Data and method

For most of recorded history, life expectancy at birth hovered between 25 and 40 years — not because people aged quickly, but because childhood mortality dominated the average. Parish registers, early censuses, and model life tables underwrite the pre-modern estimates; modern national statistics and UN-style compilations underwrite the recent ones.

The modern transformation begins in earnest around 1850 with clean water infrastructure, followed by germ theory, vaccines, antibiotics, and the broader package of public health and nutrition that raised survival at every age. The file used here stitches country-year observations across that arc so that long-run medians and cross-country gaps can be read on the same axis.

500 years of survival

Global median life expectancy from 1543 to 2015

The long-run trend has three distinct phases. A slow, stalling climb from 1543 to roughly 1850 — progress exists but is fragile and frequently reversed. A steep industrial and public-health acceleration from the mid-nineteenth century through the mid-twentieth. And a late-modern continuation in which gains are smaller in absolute years but still real, especially where child mortality has already fallen.

Notice what does not happen: the curve does not plateau after 2000. Life expectancy continues rising even in the final years of the dataset. The median path from the opening years to the close — summarized in the fact box as **33.9→73** — is the calibration line for every country comparison that follows.

Who lives longest

Top 12 countries by average life expectancy across all years

Hong Kong leads at a **75.6**-year average across the full dataset. The top dozen countries form a recognizable cluster: East Asian economies, Northern and Western European states, and a handful of high-income island and city systems. The gap between first (75.6) and twelfth (~73.5) is narrow.

The gap between twelfth and the global median (**62.4**) is enormous. Longevity leadership is a crowded club at the top and a steep drop underneath. Averaging across all years also favors places with long, well-documented modern series — a coverage effect that should temper any ranking read as eternal destiny.

The split world

Distribution of life expectancy across all country-year observations

The distribution tells the most uncomfortable story in the file. It is bimodal — two peaks, not one. One cluster sits around 35–45 years, reflecting historical and lower-income observations. Another sits in the modern high-70s. The valley between them is where average life expectancy often lands.

The median (**62.4**) and mean (**60.0**) converge near that valley — which means the summary statistic can sound reassuring while describing almost nobody's lived regime. Bimodality is the visual proof that global longevity has been two worlds sharing one chart.

The frontrunners

Life expectancy trajectories for the top-ranked countries over time

The frontrunner countries do not rise together at the same rate. Japan's trajectory is among the most striking: relatively ordinary through early modern data, then a steep climb into the global lead. Iceland and Sweden present a different profile — high from the earliest modern observations, climbing steadily rather than surging.

What the trajectory chart teaches is path dependence. Places that entered the twentieth century with strong public-health foundations stayed high. Places that industrialized later sometimes closed the gap faster in absolute years — but the membership of the top tier remains sticky once child survival and chronic-disease care are both in place.

Where longevity concentrates

Cumulative share of total life-years by country rank

The Pareto curve for life expectancy is shallower than you might expect from a deeply unequal dataset. The top countries account for a meaningful but not totalizing share of aggregate life-years. Longevity, unlike wealth, has a hard biological ceiling: no country can compound past the limits of human survival the way capital can compound without bound.

What the curve still reveals is concentration of advantage. A thin set of high-income systems occupies the long end of the distribution year after year. Closing the gap is possible — the long-run median proves that — but catching the frontrunners requires repeating their public-health and living-standard package, not wishing the ceiling higher.

What this file cannot tell you

Historical life expectancy estimates — particularly before 1800 — carry substantial uncertainty. They are reconstructed from fragmentary parish records, model life tables, and incomplete censuses. Country coverage is selective: some regions, especially parts of sub-Saharan Africa and Central Asia, have sparse pre-twentieth-century series.

Life expectancy at birth is also not the same as healthy lifespan or adult survival conditional on reaching age five. The file measures a powerful summary of mortality, not the full texture of aging, disability, or quality of life.

What to take away

The most important fact in 472 years of life expectancy data is that the trend is real. Humans have reliably extended average lifespan through sanitation, medicine, nutrition, and institutions — from a world near 30 years to a median above 73.

The second most important fact is the gap. Countries at the top of this distribution still live far longer than countries at the bottom in the same closing years. The curve shows what is possible. The spread shows what remains unfinished.

Sources

Which U.S. States Exercise the Most?

Exercise is unevenly distributed across American states — a public-health map disguised as lifestyle branding. The TidyTuesday exercise extract used here holds **52** records (states plus D.C.) with a median of **23.0** on the adults metric and a high of **32.0** in Colorado.

Mountain and New England states dominate the leaders. The charts ask how wide the spread is, how concentrated the top share is, and how adult rates relate to the men's series in the file.

Fast facts

The numbers that set the scale for this report:

52Records in the working dataset
23.0Median Adults
32.0Highest observed Adults
ColoradoTop State by Adults

Data and method

The source is the TidyTuesday release from 2018-07-17 (week16_exercise.xlsx). After cleaning, 52 rows remain.

Adults is the primary ranked metric; Men appears in the scatter. Charts are Plotly JSON with PNG fallbacks.

Western and New England States Lead Adult Exercise

Adults by State

Colorado leads at **32.0**, with Idaho at **31**, New Hampshire and the District of Columbia at **30**, and Massachusetts and Vermont at **29**. Utah and Washington continue the upper band at **28**.

The leaders share outdoor access, income mixtures, and age structures that favor measured activity — not a single policy switch.

The Top Dozen Sits Well Above the National Median

Colorado leads at 32.0 — 28.5 marks the median among the top dozen

Colorado remains first at **32.0**, while the median among the top dozen is **28.5** — five and a half points above the overall median of 23.0.

That gap is the map's main political fact: "active America" is regional, not evenly sprinkled across all 52 rows.

Most States Cluster Near the Low-20s

Median 23.0 vs mean 22.6 — the shape is relatively symmetric

The distribution's tallest bin centers near **23.5** with about **15** states, with smaller counts in both the teens and the high-20s/30s. Mean (~22.6) sits close to the median — a relatively symmetric shape for a state-level rate.

Symmetry here still allows a meaningful elite: Colorado's 32 is not a different planet, but it is a clear right-edge state.

Top States Hold a Disproportionate Share of the Aggregate

The top 5 state entries account for 36% of the aggregate adults

The Pareto curve shows the top five state entries accounting for about **36%** of the aggregate adults metric in this ranking view, with the share climbing toward 100% by the fifteenth entry.

Even in a 52-row file, the most active states carry a sizable slice of the summed activity measure.

Adult and Male Rates Move in the Same Direction

Adults vs Men

Adults versus Men across 52 rows tracks upward: states with higher adult figures also tend to post higher men's figures, with leaders reaching the low-to-mid 30s on the adult axis and into the 30s–40s on the men's axis in this extract.

The scatter's simplicity is useful: gender-specific series confirm the same state hierarchy rather than rewriting it.

What this file cannot tell you

Self-reported exercise measures vary by survey wording and year. State averages hide urban/rural and racial gaps inside borders. D.C. is not a state but sits in the file.

The 2018 TidyTuesday snapshot is not a live CDC dashboard; treat rankings as structural for this extract.

What to take away

Across 52 jurisdictions, adult exercise centers at 23.0 while Colorado reaches 32.0 and the top-dozen median hits 28.5.

Cite the regional leader cluster (Mountain West + New England) when explaining the map, and the ~36% top-five share when explaining concentration.

Sources

Data Science Learning Community. (2018). *TidyTuesday: Exercise USA*. https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2018/2018-07-17/week16_exercise.xlsx